Checkpoint One Status Report

A Comprehensive Analysis of the Indian Startup Ecosystem

Team 40

1. Meta-Information

Open Project Title	A Comprehensive Analysis of the Indian Startup Ecosystem
Group Number	40
Mentor Name	Duggineni Venkata Yugesh
Team Members	Avanith (25B1013), Shanmukha (25B0907), Tejesh (25B0905), Anirudh (25B1021)
GitHub Repository	*[https://github.com/Avanith73/CS215-Project]*

2. Progress

2.1 Datasets

Dataset 1: Startup Funding (2015–2021). Records of Indian startup investments containing investment dates, startup names, industries, cities, funding amounts, and investment stages.

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Year	2015	2016	2017	2018	2019	2020	2021	**Total**
Records	4762	1018	685	310	111	978	1209	9073

Dataset 2: Indian Unicorns (May 2025). A curated list of 118 Indian unicorn startups, including location, sector, time-to-unicorn, valuation, investors, and annual revenue.

2.2 Data Cleaning and Preprocessing

The following steps have been completed:

1. Yearly funding files were standardised to a common seven-column schema and consolidated into `merged.csv`.
2. Startup name variants were normalised. City names such as Bangalore and Bengaluru were consolidated, while cities outside India were grouped under "Abroad".
3. Empty and irrelevant rows were removed. Investment stages were mapped to ten categories: Pre-Seed/Angel, Seed Funding, Pre-Series A, Series A, Series B, Series C, Series D & Above, Private Equity, Debt/Bridge, and Undisclosed/Other.
4. Funding amounts were converted to numeric values. Valuations, annual revenues, and total funding in the unicorn dataset were converted to crores of Indian rupees.

Updated Spreadsheet: Google Sheets link

2.3 Division of Responsibilities

Data collection and cleaning were performed collaboratively. Avanith, Shanmukha, and Tejesh primarily handled Dataset 1, while Anirudh focused on the analysis of Dataset 2.

2.4 Challenges Encountered

1. **Uneven yearly coverage:** The dataset contains 4,762 records for 2015 but only 111 for 2019, potentially distorting comparisons based on raw deal counts.
2. **Inconsistent industry labels:** The 2015 Industry/Vertical entries are more elaborate than those in subsequent years, complicating sector-wise standardisation.
3. **Incomplete dates:** Where the exact day was unavailable, the first day of the corresponding month was assigned as a placeholder. These dates are approximate and unsuitable for precise daily-level analysis.

3. Refined Problem Statement

This study examines the geographical and sectoral distribution of startup investment in India between 2015 and 2021, alongside the characteristics and funding trajectories of startups that eventually attained unicorn status.

3.1 Research Question 1: Evolution of Startup Funding

How did the distribution of startup investment across Indian cities and sectors evolve between 2015 and 2021?

The analysis will compare cities' shares of disclosed funding and deal counts to assess changes in Bangalore's relative position alongside Delhi-NCR, Mumbai, and emerging hubs. It will also identify leading recipient sectors and examine shifts in their funding shares, particularly around the COVID-19 period.

3.2 Research Question 2: Characteristics of Unicorns

What funding and business characteristics are associated with Indian startups that achieve unicorn status?

The analysis will investigate the time required to reach a US$1 billion valuation, the geographical and sectoral distribution of unicorns, and variation in their reported valuations and annual revenues.

Investment stages will also be examined to understand funding trajectories as startups progress towards unicorn status. Comparing early-stage and late-stage investments may reveal common financing patterns and differences across sectors and cities.

4. Methodology

Both datasets will be analysed using descriptive statistics, comparative analysis, and visualisation.

- **Funding analysis:** Use total disclosed funding and deal counts to compare cities, sectors, and years. Transactions with undisclosed amounts will count as deals but will be excluded from monetary totals.
- **Unicorn analysis:** Examine time-to-unicorn, location, sector, valuations, revenues, and investment-stage patterns.
- **Temporal analysis:** Investigate changes in funding activity and composition around the COVID-19 period.

The study is exploratory: it aims to identify patterns and associations rather than establish causal relationships.

5. Limitations

- Funding coverage varies substantially across years, and many transactions have undisclosed amounts.
- Inconsistent labels and approximate investment dates may affect comparisons and temporal precision.
- The unicorn dataset is subject to survivorship bias because it includes only successful companies. Its financial figures may also reflect different reporting dates.

6. Next Steps

The next phase will involve city-wise, sector-wise, and year-wise funding analysis; investigation of changes around COVID-19; analysis of unicorn characteristics and investment stages; and the preparation of visualisations with appropriate consideration of data quality and coverage.